

PACKAGE INSERT

A. BRAND OR PRODUCT NAME

Isocor 2.5 mg/mL solution for injection or infusion

B. NAME AND STRENGTH OF ACTIVE SUBSTANCE(S)

1ml of solution contains 2.5 mg of verapamil hydrochloride as an active substance.

Each ampoule of 2 ml contains 5 mg verapamil hydrochloride.

Excipients: contains 17 mg of sodium under the sodium chloride form.

Full list of excipients: sodium chloride, hydrochloric acid and water for injections.

C. PRODUCT DESCRIPTION

Solution for injection or infusion.

A liquid, sterile, practically free of particles solution.

The solution after dilution is clear, colourless and practically free from particles.

D. PHARMACODYNAMICS / PHARMACOKINETICS

D.1 Pharmacodynamic properties

Pharmacotherapeutic group: Phenylalkylamine derivatives ATC-code: C08DA01

Verapamil (verapamil hydrochloride) is a calcium antagonist or slow channel inhibitor with class IV anti-arrhythmic activity. It exerts its pharmacological effects mainly by modulating the influx of ionic calcium across the cell membrane of the vessel wall as well as in myocardial cells. Verapamil hydrochloride is a phenylalkylamine derivative. Verapamil injection contain racemic mixture of l- and d-verapamil; the l-enantiomer is 10 to 18 times more potent as a calcium channel blocker than is d-enantiomer.

The principal physiologic action of verapamil is to inhibit the trans-membrane influx of extracellular calcium ions across the membranes of myocardial cells and vascular smooth muscle cells, without changing serum calcium concentrations. The exact mechanism whereby verapamil inhibits calcium ion influx is not known. The duration of action ranges from 8 to 10 hours for conventional tablets and from 12 to 24 hours for the modified release formulation, while for injection it is less than 2 hours.

By reducing intracellular calcium concentrations in cardiac and vascular smooth muscle cells, verapamil reduces myocardial contractility (causing negative inotropy) and dilates coronary and peripheral arteries. In vasospastic angina, a relaxation of coronary arteries and inhibition of coronary artery spasm improves blood flow and oxygen supply to myocardial tissues. Dilation of systemic arteries by verapamil results in a decrease in total peripheral resistance, systemic blood pressure, and the afterload of the heart (and concomitantly, lessened myocardial wall tension and reduced oxygen requirements of the myocardial tissues). Verapamil's negative inotropic effects are seldom seen in subjects with normal ventricular function. Even in patients with cardiac disease, the negative inotropic effect is usually counterbalanced by reduced afterload and cardiac output remains unchanged; only in patients with severe heart failure, decreases in cardiac output may occur.

The inhibited influx of calcium ions in cardiac tissues prolongs the refractory period and results in slowed atrioventricular nodal conduction. Normal sinus rhythm is usually not affected (some slowing may take place) except in some elderly patients or those with sick sinus syndrome. Normal atrial action potential or intraventricular conduction are not altered. The antegrade refractory period of an accessory conducting pathway may be shortened.

Verapamil may prolong the electrocardiographic PQ-interval, which is usually correlated with plasma verapamil concentrations (especially during initial verapamil treatment). Verapamil has little or no effect on the duration of the QRS complex or the QT interval.

In addition to the haemodynamic and electrophysiologic effects, verapamil has been shown to have direct cardioprotective effects during myocardial ischemia, which may also partially result from inhibition of platelet aggregation by verapamil and verapamil's antiatherogenic effects. In addition, verapamil exhibits local anaesthetic action. Inhaled verapamil has been shown to have bronchodilatory effects. Verapamil is also known to exert prostaglandin E2 antagonistic effects, which may enhance prostacyclin synthesis in the vascular endothelium.

D.2 Pharmacokinetic properties

Verapamil is widely distributed throughout the body, with a mean volume of distribution ranging from 110 to 420 litres after intravenous administration and 260 to 410 litres after oral administration. Higher distribution volumes have been observed in patients with cardiovascular, renal, or hepatic disease. Verapamil crosses the blood-brain barrier and placenta and is distributed in the breast milk. Verapamil is approximately 90% bound to plasma proteins and binding is generally independent of plasma concentration.

Verapamil is metabolised in the liver, primarily by N-dealkylation and O-demethylation. Norverapamil, the primary metabolite, may achieve plasma concentrations similar to the parent compound. There is wide interindividual variation in verapamil metabolism due to the extensive firstpass metabolism. During long-term verapamil administration, first-pass metabolism may be reduced and hepatic enzymes saturated. Verapamil exhibits bi- or triphasic elimination kinetics, with a terminal half-life of 2-8 hours following a single oral dose or after intravenous administration. After repeated doses, this increases to 4.5-12 hours. Urinary excretion accounts for 70% and fecal for 16% of verapamil elimination. Less than 4% is excreted unchanged.

The pharmacokinetics of verapamil is usually unchanged in patients with impaired renal function. The elimination of verapamil is not altered by haemodialysis, hemofiltration, or peritoneal dialysis. In elderly patients, total body clearance and volume of distribution may be reduced and peak plasma concentrations increased, suggesting a possible decrease in first-pass metabolism of verapamil. In infants, clearance is similar to that of adults but verapamil metabolism is altered.

Verapamil acts within 5 minutes of intravenous administration and in 1 to 2 hours after oral administration. Plasma concentrations of at least 100 ng/ml are usually required for acute antiarrhythmic effect. No statistically significant relationship between plasma verapamil concentrations and blood pressure reduction has been established.

E. INDICATIONS

For severe angina not responsive to oral therapy.

It may also be used in the treatment of atrial arrhythmias and supraventricular paroxysmal tachycardia. Severe hypertension and hypertensive crisis.

F. RECOMMENDED DOSAGE

Prior to intravenous administration, the solution should be visually inspected for impurities.

Usual initial intravenous dose for adults is 5 mg (one ampoule) administered slowly over a one-minute period with continuous electrocardiographic and blood pressure monitoring. If response is not adequate, the dose can be repeated in 5-10 minutes. Maintenance dosage should be adjusted individually according to clinical response.

Verapamil may also be given as an infusion, 5–10 mg (1–2 ampoules) per hour in 5% glucose or 0.9% sodium chloride solutions. The total daily intravenous dose should not exceed 100 mg (20 ampoules).

For infants, the usual intravenous dose is 0.1–0.2 mg per kg of body weight (usual single dose range, 0.75 to 2 mg or 0.3 to 0.8 ml). For children one to fourteen years of age, the usual dose is 0.1–0.15 mg per kg of body weight (usual single dose range, 2 to 5 mg or 0.8 to 2.0 ml). The dose must be administered slowly (over at least five minutes) with continuous electrocardiographic monitoring.

Injection should be stopped when desired response is obtained. If adequate response is not obtained, the dose may be repeated in 5–10 minutes.

In elderly patients, the dose should be given slowly over two minutes to minimize the risk of untoward effects.

In patients with liver dysfunction, the single dose usually does not need to be reduced but its effect may last longer. In maintenance treatment, dosage reduction is needed.

In patients with impaired renal function, dosage adjustment is usually not needed but the clinical response must be carefully monitored to prevent possible overdosage.

G. ROUTE OF ADMINISTRATION

For intravenous use only.

H. Contraindications

Verapamil injection are contraindicated in:

- Congestive (uncompensated) heart failure
- Severe left ventricular dysfunction (i.e. pulmonary wedge pressure greater than 20 mmHg or left ventricular ejection fraction less than 20–30%)
- Severe hypotension (systolic pressure less than 90 mmHg) or cardiogenic shock
- Second or third degree atrioventricular block (except in patients with a functioning pacemaker)
- Sick sinus syndrome (except in patients with a functioning artificial ventricular pacemaker)
- Marked bradycardia
- Atrial flutter or atrial fibrillation in conjunction with an accessory conducting pathway (e.g. Wolff-Parkinson-White and Lown-Ganong-Levine syndromes)
- Digitalis-intoxication
- Hypersensitivity to verapamil hydrochloride or any of the excipients
- Wide-complex ventricular tachycardia
- Concurrent administration of intravenous beta-adrenergic blocking agents

I. WARNINGS AND PRECAUTIONS

Verapamil may cause asymptomatic first-degree atrioventricular block and transient bradycardia, sometimes accompanied by nodal escape rhythms. Prolongation of the electrocardiographic PQ-interval is associated with plasma verapamil concentrations, especially at the initiation of treatment.

Patients with hypertrophic cardiomyopathy are at increased risk of adverse cardiac effects. The incidence of cardiac adverse effects is also increased in patients concurrently treated with beta adrenergic blocking agents.

Verapamil clearance is decreased and its elimination half-life prolonged in elderly patients. Special care is required in using verapamil in neonates and infants, as they may be more susceptible to verapamil-induced arrhythmias.

Verapamil dosage should be reduced in patients with hepatic disease.

Although dosage adjustment in patients with renal impairment is usually not necessary, such patients must be monitored closely. Verapamil treatment is continued during dialysis.

In some patients, concurrent administration of verapamil and alcohol (ethanol) can lead to impaired elimination of ethanol.

Verapamil should be used with caution in patients with mild to moderate heart failure; signs and symptoms of heart failure should be controlled with cardiac glycosides and/or diuretics before initiating verapamil treatment. Verapamil dosage may need to be reduced in patients with attenuated neuromuscular transmission (e.g., diseases affecting the neuromuscular transmission, use of neuromuscular blocking agents during anesthesia).

Dosage should be reduced or medication discontinued and appropriate treatment instituted, if congestive heart failure or conduction disturbances develop.

Liver function should be periodically monitored during verapamil treatment.

Intravenous verapamil should be initially used only in a setting where resuscitation capabilities are available. All patients receiving intravenous verapamil should be under electrocardiographic and hemodynamic monitoring.

J. INTERACTIONS WITH OTHER MEDICAMENTS

Pharmacodynamic interactions with or without a pharmacokinetic component

Verapamil interacts pharmacodynamically with beta-adrenergic blocking agents. Concurrent use may increase the risk of heart failure, arrhythmias, and severe hypotension. In addition, a pharmacokinetic interaction (of only limited clinical significance) has been demonstrated with at least metoprolol, atenolol, and propranolol, resulting in increased plasma concentrations of these beta-blockers.

Adverse pharmacodynamic interactions with amiodarone, disopyramide, flecainide, prazosin and clonidine may occur, as well as reduced clearance and higher plasma concentrations, when administered concomitantly with verapamil. The risk of cardiotoxicity increases.

Additive negative inotropic and hypotensive effects may occur during use of inhalation anaesthetics.

Enhanced antiplatelet effects may result from concurrent administration of verapamil with acetylsalicylic acid (aspirin), while acetylsalicylic acid may decrease the antihypertensive effect of verapamil.

Coadministration of intravenous dantrolene and verapamil may result in hyperkalaemia and severe cardiovascular depression.

Variable effects of coadministration of verapamil on plasma lithium concentrations have been observed. Increased neurotoxicity of lithium may result from combined treatment.

The effect of neuromuscular-blocking agents employed in anaesthesia may be potentiated.

Pharmacokinetic interactions

Verapamil has been shown to be metabolised predominantly via the cytochrome P450 3A4 (CYP3A4) system. It is also an inhibitor of CYP3A4. Accordingly, the drugs that inhibit or induce this enzyme may affect the metabolism of verapamil. On the other hand, verapamil may affect the metabolism of other drugs which are metabolised via CYP3A4. Verapamil is also a substrate and an inhibitor of P-glycoprotein.

Effect of other drugs on verapamil

Calcium salts may antagonise the pharmacological response to verapamil.

Metabolism of verapamil can be inhibited by coadministration of macrolide antibiotics (erythromycin, clarithromycin) and grapefruit juice. Concurrent administration of cimetidine or HIV protease inhibitors such as ritonavir may increase the plasma concentration of verapamil which can cause toxic effects.

Effect of verapamil on other drugs

Verapamil increases the plasma levels of antiarrhythmic agents, for example quinidine, which may lead to cardiotoxicity.

Verapamil increases the plasma concentrations of digoxin, and to a lesser extent, of digitoxin.

Hepatic metabolism of simvastatin, atorvastatin and lovastatin is inhibited by verapamil. This may cause significantly increased plasma levels of those drugs. If taken concomitantly, a small dose of statin must be used and symptoms of rhabdomyolysis and hepatic damage must be closely monitored.

Increased plasma concentrations of carbamazepine, theophylline, sirolimus and tacrolimus have been observed during concurrent administration of verapamil.

K. STATEMENT ON USAGE DURING PREGNANCY AND LACTATION

Verapamil crosses the placenta and can be detected in umbilical vein blood at delivery. Rapid intravenous injection of verapamil may cause maternal hypotension resulting in fetal hypoxia.

Verapamil has occasionally been used during treatment of premature labour; prolongation of the duration of labour or increased need for obstetric interventions has not been reported. Nevertheless, no controlled studies on the safety of verapamil during pregnancy or labour exist and, therefore, risk–benefit should be thoroughly weighed and verapamil used only if expected benefits outweigh possible risks to the fetus.

Verapamil is excreted in human breast milk. Verapamil concentrations in human milk may be variable but similar concentrations to those found in maternal plasma have been reported. Available data indicate that adverse effects in infants during breast-feeding are unlikely. However, it is recommended that nursing should be continued only if it is essential for the welfare of the patient.

L. ADVERSE EFFECTS / UNDESIRABLE EFFECTS

The majority of adverse reactions to verapamil are extensions of its vasodilatory action and negative inotropic properties, and are generally more common after intravenous administration. Serious adverse reactions are uncommon when verapamil treatment is initiated with upward dose titration within the recommended single and total daily dose. Other adverse effects (related to vasodilatation) include headaches, flushing, dizziness, and ankle edema. Gingival hyperplasia may develop in long-term treatment. These minor adverse effects rarely necessitate discontinuation of verapamil treatment. Major adverse effects, that often require dosage reduction or drug discontinuation, are infrequent. Those occur mainly in patients with left ventricular dysfunction or nodal disease or those on beta-adrenergic blocking agents, and include bradycardia, atrioventricular block, hypotension, congestive heart failure, or even asystole.

The following adverse events have been reported during verapamil hydrochloride treatment:

Common

Cardiac: Bradycardia, cardiac failure

Vascular: Hypotension, swelling of ankles

Gastrointestinal: Constipation, dyspepsia, nausea

Nervous system: Dizziness, headache, lethargy

Skin: Rash, exanthema, urticaria, pruritus

General: Fatigue

Others

Cardiac: First-, second- or third degree atrioventricular block, palpitation, asystole

Vascular: Flushing, intermittent claudication

Gastrointestinal: Dry mouth, gingival hyperplasia, diarrhoea, ileus

Nervous system: Paresthesias, muscle tremor, syncope

Psychiatric: Confusion, depression, sleep disturbances, nightmares

Eye: Visual disturbances

Ear: Tinnitus

Renal and urinary: Pollakisuria

Reproductive system and breast: Impotence, spotty menstruation, gynaecomastia

Endocrinological: Hyperprolactinaemia (with associated galactorrhoea)

Musculoskeletal: Myalgia, arthralgia

Skin: Angioneurotic oedema, purpura, bruising, erythema multiforme, Stevens-Johnson syndrome, and photosensitivity

Respiratory: Bronchospasm, laryngospasm

Investigations: Elevation of liver enzyme levels

General: Chest pain

M. OVERDOSE AND TREATMENT

Overdose with verapamil is a medical emergency that can be fatal. Central nervous system features of overdose include drowsiness, confusion, and rarely seizures, even coma secondary to cardiovascular collapse may occur. Nausea, vomiting, metabolic acidosis, and secondary hyperglycemia may also occur. Hypotension is the most common cardiovascular finding; bradycardia, atrioventricular block, sinus arrest with nodal escape rhythms, and asystole may occur.

Treatment of overdose should be individualized based on the clinical situation and the judgment and experience of the treating physician.

Treatment of cardiovascular effects is supportive and symptomatic; intravenous calcium, atropine or isoprenaline, and glucagon may be helpful. Verapamil is not removed by hemodialysis. Continuous cardiac monitoring is recommended.

N. EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

Depending on individual sensitivity to verapamil, ability to drive and use machines may be impaired, especially at the initiation of treatment. During verapamil treatment, blood alcohol (ethanol) concentrations may be higher and the intoxicating effect of alcohol longer than normally expected.

O. INSTRUCTIONS FOR USE

Verapamil hydrochloride should be given as a slow intravenous injection over at least a two minute period of time under continuous electrocardiographic and blood pressure monitoring.

Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit.

Intravenous infusions of 100 ml:

1 ampoule of 2 ml (5 mg) of Isocor 2.5 mg/mL solution for injection or infusion - 2 ml in 100 ml of the relevant solution for infusion. The final concentration of the infusion solution is 49 µg/ml verapamil hydrochloride.

Use only if solution is clear and vial seal is intact.

Unused amount of solution should be discarded immediately following withdrawal of any portion of contents.

For stability reasons this product is not recommended for dilution with sodium lactate injection, USP in polyvinylchloride bags.

Admixing intravenous verapamil hydrochloride with albumin, amphotericin B, hydralazine hydrochloride or trimethoprim and sulfamethoxazole should be avoided.

Verapamil hydrochloride will precipitate in any solution with a pH above 6.0.

Verapamil should not be taken with grapefruit juice (see Section J, Interactions).

Isocor is physically and chemically compatible with Ringer solution, 5% Glucose or 0.9% NaCl and may be applied concomitantly with each of them.

From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user.

P. INCOMPATIBILITIES

Incompatible with alkaline solutions.

In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products.

Q. STORAGE CONDITIONS

Store below 25°C.

Store in the original package in order to protect from light.

Do not freeze!

Isocor 2.5 mg/mL solution for injection or infusion should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user after reconstitution and further dilution

R. DOSAGE FORMS AND PACKAGING AVAILABLE

The drug product is dosed in ampoules of colourless glass, Type I with capacity of 2 ml, with a marking for ampoule opening. A self-adhesive label is fixed on each ampoule.

10 (ten) ampoules are placed in a blister strip of PVC film.

1 (one) blister strip is placed in a carton. A package leaflet with instructions for use is placed in each carton.

S. NAME AND ADDRESS OF MANUFACTURER

SOPHARMA AD
16 Iliensko Shosse Str.
1220 Sofia
Bulgaria

T. PRODUCT REGISTRATION HOLDER

Rigel Pharma Sdn Bhd
3rd Floor, Apex Healthcare Building.
No. 2, Jalan SS 13/5, 47500 Subang Jaya

Selangor, Malaysia

U. DATE OF REVISION OF PI

December 2022

Note:

Font size in actual package insert: 7 (as measured in Times New Roman)